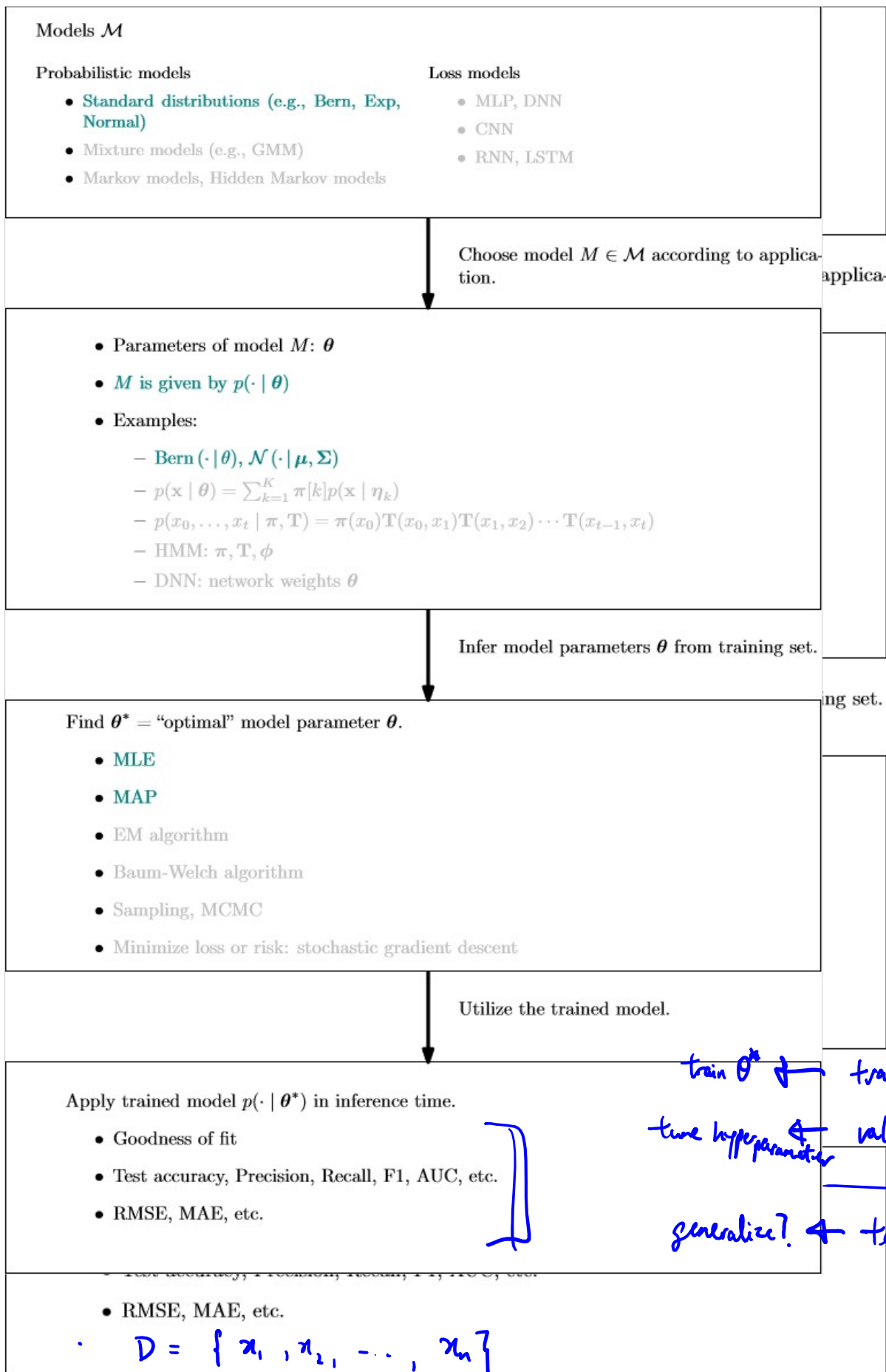


Week 2

Saturday, August 15, 2026 7:25 PM



overview
week 2



$x_i \sim \text{Bern}(\cdot | \theta)$ i.i.d. independent and identically distributed

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likelihood $p(D | \theta) = \prod_{i=1}^n p(x_i | \theta) = \prod_{i=1}^n \text{Bern}(x_i | \theta)$ $x_i \in \{0, 1\}$

$\uparrow$ independence

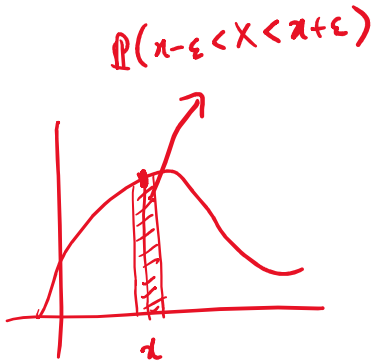
$$= \prod_{i=1}^n \theta^{x_i} (1-\theta)^{1-x_i}$$

$$= \theta^{\sum_{i=1}^n x_i} (1-\theta)^{n - \sum_{i=1}^n x_i}$$

$$= \theta^{N_1} (1-\theta)^{N_0}$$

$$N_0 + N_1 = n$$

$$= f^n \text{ of } \theta \quad l(\theta)$$



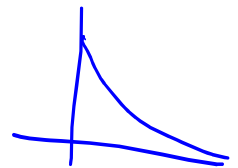
$x_i \sim \text{Exp}(\cdot | \lambda)$ i.i.d.

$$p(D | \lambda) = \prod_{i=1}^n p(x_i | \lambda)$$

$$= \prod_{i=1}^n \lambda e^{-\lambda x_i}$$

$$= \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

$$, x_1, \dots, x_n \geq 0$$



Maximum Likelihood Estimate (MLE)

$$\theta_{ML} = \underset{\theta}{\text{argmax}} \underset{\text{likelihood}}{p(D | \theta)}$$

$$\begin{aligned} \arg \max_{\theta} p(D|\theta) &\equiv \arg \max_{\theta} \log p(D|\theta) && \log - \text{increasing func.} \\ \parallel &\parallel && a < b \\ \prod_{i=1}^n p(x_i|\theta) &\sum_{i=1}^n \log p(x_i|\theta) && \log a < \log b \end{aligned}$$

diff: product chain rule

$$\sum_{i=1}^n \frac{d}{d\theta} \log p(x_i|\theta)$$

$$\max_{\theta} h(\theta) \quad / \quad \text{if differentiable: } \frac{d}{d\theta} h(\theta) = 0 \Rightarrow \theta^*$$

$$\frac{d^2}{d\theta^2} h(\theta) < 0$$

\ not differentiable. cannot do $\frac{d}{d\theta}$.

$$p(D|\theta) = \theta^{N_1} (1-\theta)^{N_0}$$

$$\log p(D|\theta) = N_1 \log \theta + N_0 \log(1-\theta) \quad \boxed{\begin{matrix} \log \\ \ln \end{matrix}}: \text{natural logarithm} \quad \log_{10}$$

$$\frac{d}{d\theta} \log p(D|\theta) = \frac{N_1}{\theta} - \frac{N_0}{1-\theta} = 0$$

$$\theta_{ML} = \frac{N_1}{N_0 + N_1} = \frac{N_1}{n}$$

$$p(D|\theta) = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

$$\log p(D|\theta) = n \log \lambda - \lambda \sum_{i=1}^n x_i$$

$$\frac{d}{d\lambda} : \quad \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0$$

$$\lambda_{ML} = \frac{n}{\sum x_i}$$

$$\lambda_{ML} = \frac{1}{\sum_{i=1}^n x_i}$$

i.i.d. $\sim \theta_0$

SLLN

$$\max_{\theta} \frac{1}{n} \log p(D|\theta) = \frac{1}{n} \sum_{i=1}^n \underbrace{\log p(x_i|\theta)}_{\text{i.i.d.}} \xrightarrow{\text{SLLN}} \mathbb{E}_{\theta_0}[\log p(x|\theta)]$$

$$\max_{\theta} \mathbb{E}_{\theta_0}[\log p(x|\theta)]$$

$$\theta^* = \theta_0 ?$$

$$\min_{\theta} \left\{ -\mathbb{E}_{\theta_0}[\log p(x|\theta)] + \underbrace{\mathbb{E}_{\theta_0}[\log p(x|\theta_0)]}_{\text{independent of } \theta} \right\}$$

$$= \mathbb{E}_{\theta_0} \left[\log \frac{p(x|\theta_0)}{p(x|\theta)} \right] = D(p(\cdot|\theta_0) \parallel p(\cdot|\theta))$$

$$\min_{\theta} D(p(\cdot|\theta_0) \parallel p(\cdot|\theta))$$

$$\geq 0 \quad "=" \quad \text{iff } \theta = \theta_0$$

Cross-entropy

K classes $x \rightarrow y \in \{1, 2, \dots, k\}$

$$q(\cdot | x, \theta) \quad \begin{bmatrix} 1 \\ 2 \\ \vdots \\ k \end{bmatrix}$$

$$D = \{ (x_i, y_i) : i=1, \dots, n \}$$

$$p(x, y) = p(y|x) p(x)$$

$$\log p(D|\theta) = \sum_{i=1}^n \log p(x_i, y_i | \theta)$$

$$= \frac{n}{2} \log \left(\frac{1}{2\pi} \right) - \frac{n}{2} \log \left(\frac{1}{2} \right) - \frac{n}{2} \log \left(\frac{1}{2} \right)$$

$$= \sum_{i=1}^n \log q(y_i | x_i, \theta) + \sum_{i=1}^n \log p(x_i | \theta)$$

$$(-n) \left(-\frac{1}{n} \sum_{i=1}^n \left(\sum_{k=1}^K \mathbb{1}\{y_i=k\} \log q(k | x_i, \theta) \right) \right)$$

$\frac{\mathbb{1}\{y_i=k\}}{n}$
 $\xrightarrow{n \rightarrow \infty} p(y_i=k)$

$$\log q(\underline{y_1=2} | x_1, \theta) + \log q(y_2=1 | x_2, \theta) \\ + \log q(\underline{y_3=2} | x_3, \theta) + \log q(y_4=1 | x_4, \theta)$$

$$\left(\log q(y_2=1 | x_2, \theta) + \log q(y_4=1 | x_4, \theta) \right) \\ + \left(\log q(y_1=2 | x_1, \theta) + \log q(y_3=2 | x_3, \theta) \right)$$

Linear Regression

$$y = \vec{w}^T \vec{x} + \underset{\substack{\uparrow \\ \text{noise}}}{\varepsilon}$$

$$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{bmatrix} \quad \vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_D \end{bmatrix}$$

$$= \boxed{w_1 x_1 + w_2 x_2 + \dots + w_D x_D} + \varepsilon \quad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

$$y | \vec{x} \sim \mathcal{N}(w^T x, \sigma^2)$$

$\uparrow$
obs. feature

$$\mathcal{N}(y | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y-\mu)^2}$$

$$\log p(\mathcal{D} | \{\vec{x}_i\}, w) = \sum_{i=1}^n \log p(y_i | \vec{x}_i, w)$$

$$\log p(\mathcal{D} | \{\vec{x}_i\}, w) = \sum_{i=1}^n \log p(y_i | \vec{x}_i, w)$$

$$= \sum_{i=1}^n \log N(y_i | w^T \vec{x}_i, \sigma^2)$$

$$\vec{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$\Phi = \begin{bmatrix} \vec{x}_1 \\ \vdots \\ \vec{x}_n \end{bmatrix}$$

$$= \sum_{i=1}^n \left(-\frac{1}{2\sigma^2} (y_i - w^T \vec{x}_i)^2 \right) + \text{const}$$

$$= -\frac{1}{2\sigma^2} \|\vec{y} - \Phi \vec{w}\|^2 + \text{const}$$

$$L^2\text{-norm}$$

$$\|u\|^2 = \sum_{i=1}^n u_i^2$$

$$MLE: \min_{\vec{w}} \|\vec{y} - \Phi \vec{w}\|^2$$

$$\begin{array}{l} \varphi_1(\vec{x}) \\ \varphi_2(\vec{x}) \\ \vdots \\ \varphi_m(\vec{x}) \end{array} \quad \begin{array}{l} \sum_{i=1}^m w_i \varphi_i(\vec{x}) \\ \sum_{i=1}^D w_i \vec{x}[i] \end{array}$$

$$\Phi = \begin{bmatrix} \varphi_1(\vec{x}_1) & \varphi_2(\vec{x}_1) & \dots & \varphi_m(\vec{x}_1) \\ \varphi_1(\vec{x}_2) & \varphi_2(\vec{x}_2) & \dots & \varphi_m(\vec{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \varphi_1(\vec{x}_n) & \varphi_2(\vec{x}_n) & \dots & \varphi_m(\vec{x}_n) \end{bmatrix}$$

design matrix

$$\min_w \|\vec{y} - \Phi \vec{w}\|^2$$

$$\|u\|^2 = u^T u$$

$$[u_1 \dots u_D] \begin{bmatrix} u_1 \\ \vdots \\ u_D \end{bmatrix}$$

$$(\vec{y} - \Phi \vec{w})^T (\vec{y} - \Phi \vec{w})$$

$$= \underbrace{\vec{w}^T \Phi^T \Phi \vec{w}} - 2 \vec{y}^T \Phi \vec{w} + \text{const}$$

$$= \underbrace{\vec{w}^T \Phi^T \Phi \vec{w}}_A - 2 \underbrace{\vec{y}^T \Phi \vec{w}}_{a^T} + \text{const}$$

$$\frac{\partial}{\partial \vec{w}} : \quad 2 \Phi^T \Phi \vec{w} - 2 \Phi^T \vec{y} = 0$$

Assuming $\Phi^T \Phi$ is invertible,

$$\vec{w}_{ML} = (\Phi^T \Phi)^{-1} \Phi^T \vec{y}$$

MAP

$$\underbrace{p(\theta | \mathcal{D})}_{\text{posterior}} \propto \underbrace{p(\mathcal{D} | \theta)}_{\text{likelihood}} \underbrace{p(\theta)}_{\text{prior}}$$

$$\max_{\theta} p(\theta | \mathcal{D})$$

$$\log p(\theta | \mathcal{D}) = \underbrace{\log p(\mathcal{D} | \theta)}_{\text{log likelihood}} + \underbrace{\log p(\theta)}_{\text{log prior}} + \text{const}$$

$$p(\mathcal{D} | \theta) = \theta^{N_1} (1-\theta)^{N_0} \quad \text{likelihood}$$

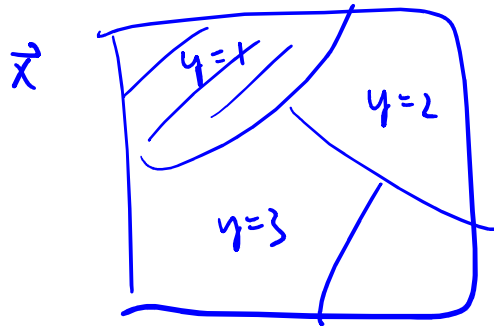
$$p(\theta) = \text{Beta}(\theta | a, b) \propto \theta^{a-1} (1-\theta)^{b-1}$$

$$\begin{aligned} \log p(\theta | \mathcal{D}) &= \log p(\mathcal{D} | \theta) + \log p(\theta) + \text{const} \\ &= \log(\theta^{N_1} (1-\theta)^{N_0}) + \log(\theta^{a-1} (1-\theta)^{b-1}) + \text{const} \\ &= N_1 \log \theta + N_0 \log(1-\theta) + (a-1) \log \theta + (b-1) \log(1-\theta) \\ &= (N_1 + a - 1) \log \theta + (N_0 + b - 1) \log(1-\theta) \end{aligned}$$

$$\frac{d}{d\theta} : \quad \frac{N_1 + a - 1}{\theta} - \frac{N_0 + b - 1}{1-\theta} = 0$$

$$\frac{d}{d\theta} : \frac{N_1 + a - 1}{\theta} - \frac{N_0 + b - 1}{1 - \theta} = 0$$

$$\theta_{\text{MAP}} = \frac{N_1 + a - 1}{n + a + b - 2}$$



$$\theta_{\text{ML}} = \frac{N_1}{n}$$

$$f(\vec{x}) = \begin{cases} 1 & \text{if } \vec{x} \in R_1 \\ 2 & \text{if } \vec{x} \in R_2 \\ \vdots & \end{cases}$$

$$\min \sum_{y \neq f(\vec{x})} \mathbb{P}(Y=y | \vec{x}) = 1 - \underbrace{\mathbb{P}(Y=f(\vec{x}) | \vec{x})}_{\text{posterior}}$$

classification error

MNIST

$$\square \rightarrow \vec{x} = [\]$$